

Computational Image Analysis of Magnetic Domains in Bi:YIG Thin Films

Author: Joseph Burton

Faculty Advisor: Professor Bellave Shivaram

Introduction

Magnetic domains are areas within magnetic materials where all the magnetic moments of the particles in that area align (1). This phenomenon arises in materials due to the tendency of matter to minimize its energy; magnetic materials seek the smallest energy state possible, and by splitting the material into many different domains, the material is able to minimize the external magnetic field it creates, which in turn minimizes the energy level stored in the field (1). The Faraday effect, which causes a polarizing rotation that is proportional to the magnetic field in the direction of the light propagation, is one way these magnetic domains can be imaged directly with the use of polarizers and analyzers (2). The Faraday effect causes a rotation of light as it passes through a magnetic material, and can be described with equation 1, where β is the

$$\beta = \nu B d \quad \text{(Equation 1)}$$

angle of rotation, ν is the Verdet constant which determines the amount of rotation (which is dependent on the material), B is the magnetic flux density in the direction the light is traveling, and d is the path length through which the magnetic field and the light interact (2). By creating a set up with a magnetic sample between two polarizers, a diffuse light source on one side, and a camera on the other side, light from the source will enter the first polarizer and then pass through the sample. The magnetic sample, with up and down magnetic domains and with some specific properties and Verdet constant, will rotate this polarized light by some angle and allow light to pass through the domains differently (3). This light then passes through the analyzer (which is tuned

to yield the maximum contrast between light and dark samples) and into the camera.

The camera will thus be able to see the difference between the magnetic domains of the sample, and we are therefore able to directly image the sample's magnetic domains.

The sample chosen for this lab was Bi:YIG: Bismuth doped Yttrium Iron Garnet, which forms magnetic domains easily, has a large Verdet constant in the visible spectrum, has a high Curie temperature so its able to retain its magnetic properties for convenient experimental ambient temperatures, and has a preferred magnetization direction that is perpendicular to the sample (3). The objective of the research we conducted was to utilize the Faraday Effect to image the magnetic domains in Bi:YIG thin film samples directly, and to explore and quantify the images of the magnetic domains in the samples under different external magnetic fields.

Methods and Experimental Procedure

Physical Setup

As the magnetic domains in the sample used are miniscule and unable to be seen with the naked eye, the domain images were taken through a microscope. A polarizing microscope was employed, operating at around 30x magnification, with a Swift Imaging camera eyepiece (3). In order to control the magnetic field, a Helmholtz coil attached to a power supply was placed around the sample, and in order to use any

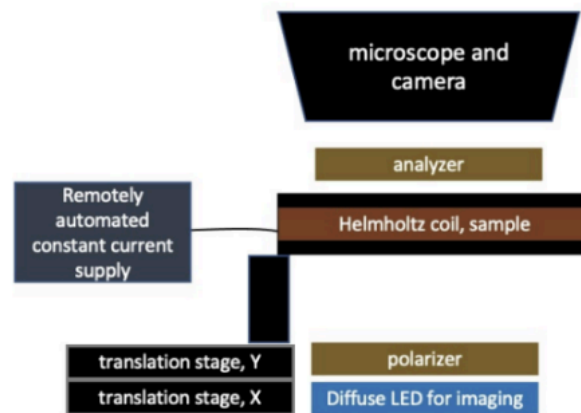


Figure 1. Diagram showing data taking setup (3).

function of voltage versus time the power supply was connected to a Waveform generator. A metal shield was placed around the sample to help control for the Earth's magnetic field (3). We placed a diffuse light source (an LED running off of a power supply with a diffuser on top) underneath the set up to illuminate the sample. We placed a polarizer between the diffuser and the sample, and we placed an analyzer between the sample and the microscope lens. These were tuned to yield the maximum contrast between the light and dark domains in the images. A diagram of the setup can be seen in figure 1 and an image of the actual setup in figure 2. The specific desired magnetic field changes to obtain data during different trials will be discussed later, but in general

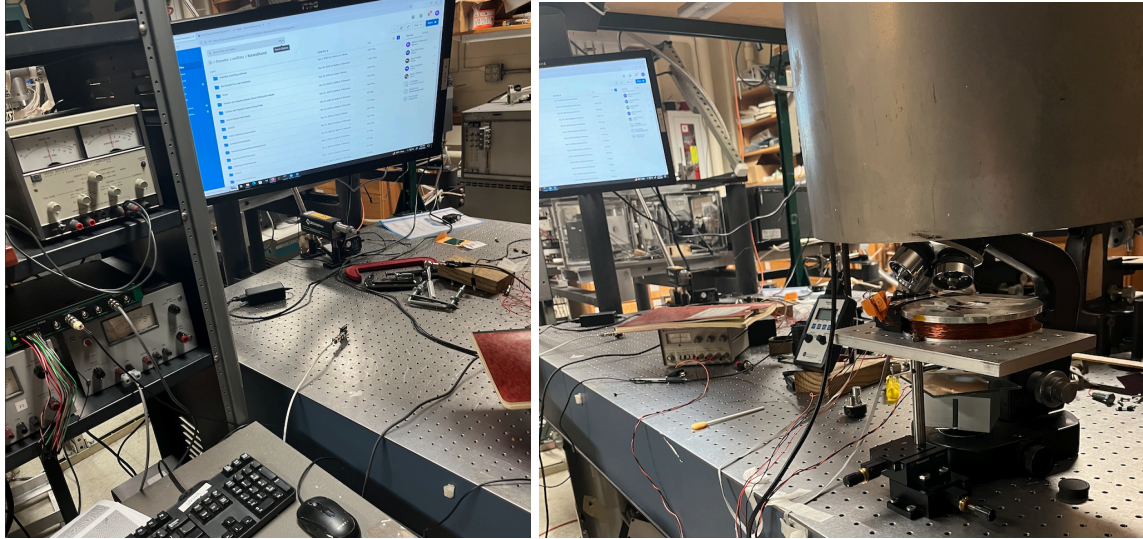


Figure 2. Photos of experimental setup. Left: computer connected to equipment, Waveform generator. Right: Helmholtz coil + sample, metal magnetic shield, microscope, gaussmeter, LED power supply.

the microscope was manually focused, and images were either taken by hand (through the Swift imaging software) or by autocapture, a program on the Swift imaging software

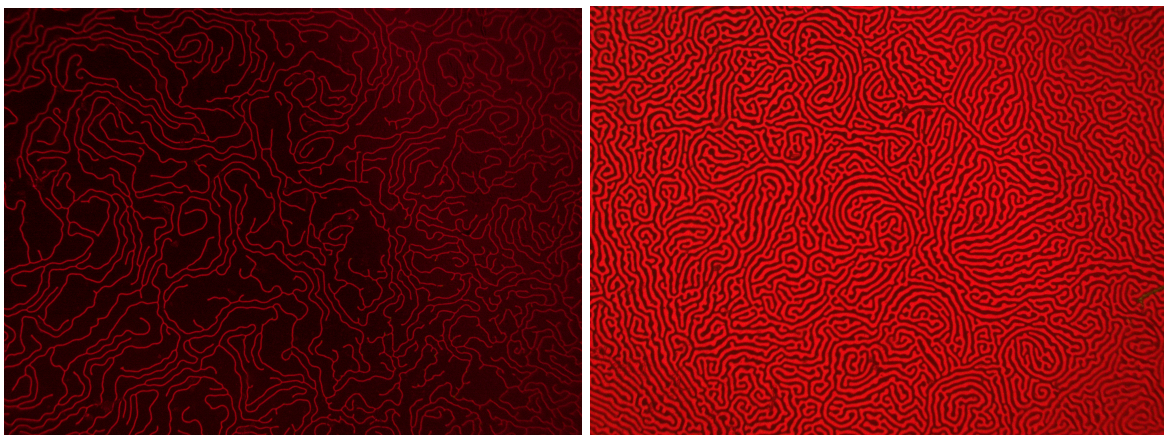


Figure 3. Example images of domains taken with experimental setup.

that allows for programmed capture. When using autocapture, the number of pictures and wait time between pictures was synchronized with the voltage input (which controlled the magnetic field) from the waveform generator to allow for images to be taken autonomously after being set up. We had two samples we could use, a red sample and a green sample, however the red sample in general had much clearer images and less imperfections so all of the following analysis will focus on the red sample's results.

Note on Image Analysis and Threshold Value

To be able to analyze the magnetic domain images, it is necessary to distinguish between different pixels in an image, and thus it's important to understand how images are stored in computers. In png images, which were used in all of the image analysis techniques, an image is broken down into a 2D array of pixels, and of these pixels is assigned three values, an R, a G, and a B value, which determine the color of the pixel. Our analysis used grayscale images; to convert a pixel from color to grayscale, that pixel's R, G, and B values (which are integers from 0-255) are averaged, and this average becomes the new R value, the new G value, and the new B value. So, turning a color pixel into a grayscale pixel essentially reduces a pixel's 3 RGB values to a single RGB value. Thus, a grayscale pixel's RGB value can be compared to another to determine which one is darker and which one is brighter. The images taken by the Swift camera were color dng images, so in order to convert these images to greyscale pngs, they were first converted to color pngs by either Image Magick or Apple's Preview app. Then, these color png images were converted to greyscale through the use of batch

processing in ImageJ 2. In the domain images, there are two colors that are dominant, which I think of as the dark region and light region. In order to distinguish between these two, a threshold value is needed. The way I found this was simple; first I took a sample of pixels and gathered their grayscale values. Then I sorted this list with respect to darkness and found the average of an early element and a late element in the list (the early element ranges from 1st to 5th percentile and the late element ranged from 95th to 99th percentile; the early and late elements were evenly spaced). These percentiles were used as an attempt to get rid of outliers: because of imperfections in the sample, some spots on the screen tend to be blacked out, which causes a darker pixel than the domains would yield otherwise, so the above method was implemented rather than using the darkest and lightest pixel values and averaging those. This was how the critical threshold value was found, which is used in both of the following algorithms.

Domain Thickness

Computational Methods

In the domain images, it can be seen that at different magnetic fields, the thickness of the domain lines in the images changes. In order to quantify this effect, I wrote an algorithm to find the average domain thickness of an image in Python. The algorithm seeks to find an average value for the dark and light domain thickness in pixels across the entire picture. First, random pixels were checked until I found one that was the desired color. Next, from that pixel, I extend counters both ways across four distinct lines; with the original pixel being at (0,0), the lines seen in equations two

$$y = x \quad y = -x \quad y = 0 \quad x = 0 \quad (\text{Equations 2-5})$$

through five are all explored. The algorithm counts how long each line is in pixels and cuts off the count when a value across the dark/light threshold value is found. The path

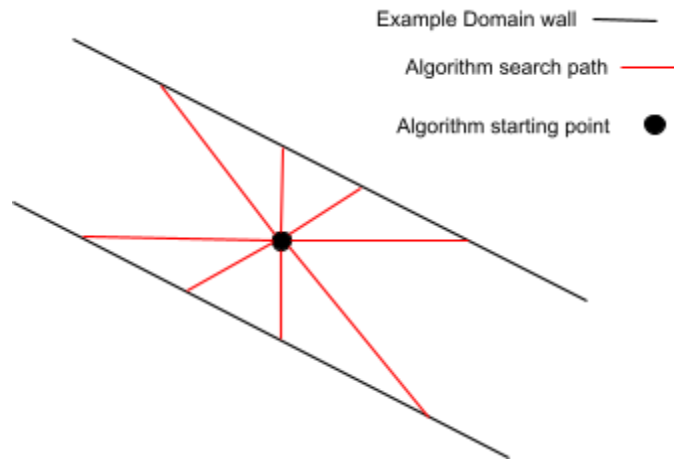


Figure 4. Domain thickness algorithm diagram. Note the algorithm search lines are always in the depicted directions, while the domain walls can be in any orientation (but are assumed to be parallel in the algorithm).

that is the shortest of these four is chosen as the approximate domain width at that location. This process is then repeated for a number of preset repetitions (the nominal iteration count was 30000, however the actual count often fell below this due to some domain lines being located on the border of the image, of which the values were discarded) and then all of these domain samples are used to find an average and standard deviation for the dark and light domain widths. This entire process is then repeated for the other color domain width, so both dark and white domains are explored and documented. A few things cause errors in the domain width calculated: one, the

fact that only four directions are explored, and two, the fact that images are broken into an array of discrete pixels. As domains appear in random orientations, there's no reason why any domain line will necessarily align perfectly with the directions that are checked in the algorithm. My algorithm checks for only those four directions, thus it will yield the shortest distance checked but this doesn't necessarily correspond with a line perpendicular to the wall of the domain line. The true value would follow equation 6, where true is the actual domain thickness, found is the measured domain thickness,

$$\text{true} = \text{found} * \cos(\theta) \quad (\text{Equation 6})$$

and θ is the angle between the shortest test line and the true thickness line (a line passing through the original test point perpendicular to the domain walls). Following this, the relative error, E, can be seen in equation 7. Taking the average value of this

$$E = | (\text{found} - \text{true}) / \text{true} | \quad (\text{Equation 7})$$

over the range of possible intervals results in equation 8, where $\Delta\theta$ is $\pi/4$ and the

$$E_{\text{avg}} = 1 / (\Delta\theta) * \int | (\text{found} - \text{true}) / \text{true} | * d\theta \quad (\text{Equation 8})$$

integral is the definite integral from 0 to $\pi/4$. Using equation six to substitute and then

$$E_{\text{avg}} = 1 / (\Delta\theta) * \int | (1 - \cos(\theta)) / \cos(\theta) | * d\theta \quad (\text{Equation 9})$$

simplifying, equation 9 is reached, where $\Delta\theta$ is still $\pi/4$ and the integral is still from 0 to $\pi/4$. Evaluating equation 9 yields E_{avg} = about 0.03975, or an average error of about 3.975%. Equation 10 can be used to find the true value from the found value, finally resulting in

$$\text{found} = \text{true} * (1 + E_{avg}) \quad (\text{Equation 10})$$

equation 11. The algorithm multiplies the final measured average value by this

$$\text{true} = \text{found} / (1 + E_{avg}) = \text{found} * 0.96176565822 \quad (\text{Equation 11})$$

correction factor, 0.96176565822, to obtain the most accurate results.

Results

The width was explored for a significant portion of the data taken; I will discuss two sets of this data, the hysteresis completed for the red sample and the step data completed for the red sample. The step data set was collected by annealing the sample multiple times to get the best, most ordered domains and an image with the smallest grains, then incrementing the field upwards from zero gauss until 50 gauss was reached. The steps had different magnitudes at different points in the trial. This whole progress was repeated but for the negative direction; started at zero gauss, incremented the magnetic field until -50 gauss was reached. After each step, an image

was taken and the magnetic field, measured by a Gaussmeter, was recorded. These data sets were performed and recorded manually.

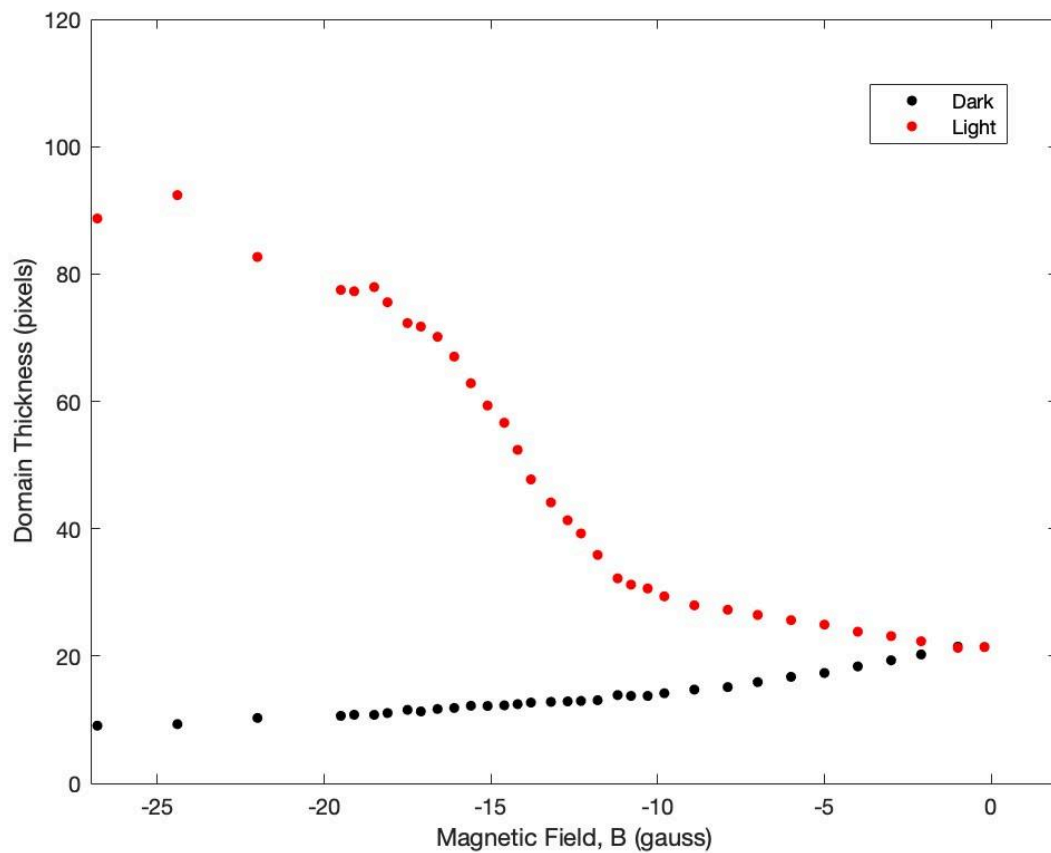


Figure 5. Domain thickness versus magnetic field.

As can be seen in figures 5 and 6, for both sets of data, the domain thickness of the dark and light regions start out somewhere close together and then diverge as the trial continues. This makes sense, as in the images it is easily noticeable that one color of domains remains as the normal thin lines but become fewer in number and less densely packed, while the other color's domain lines essentially disappear and that color becomes the background. The domain thickness results seem to break down at around

± 25 gauss in the step data (not shown in the graphs). Here the positive dark thickness starts rapidly decreasing, and the negative light domain thickness becomes

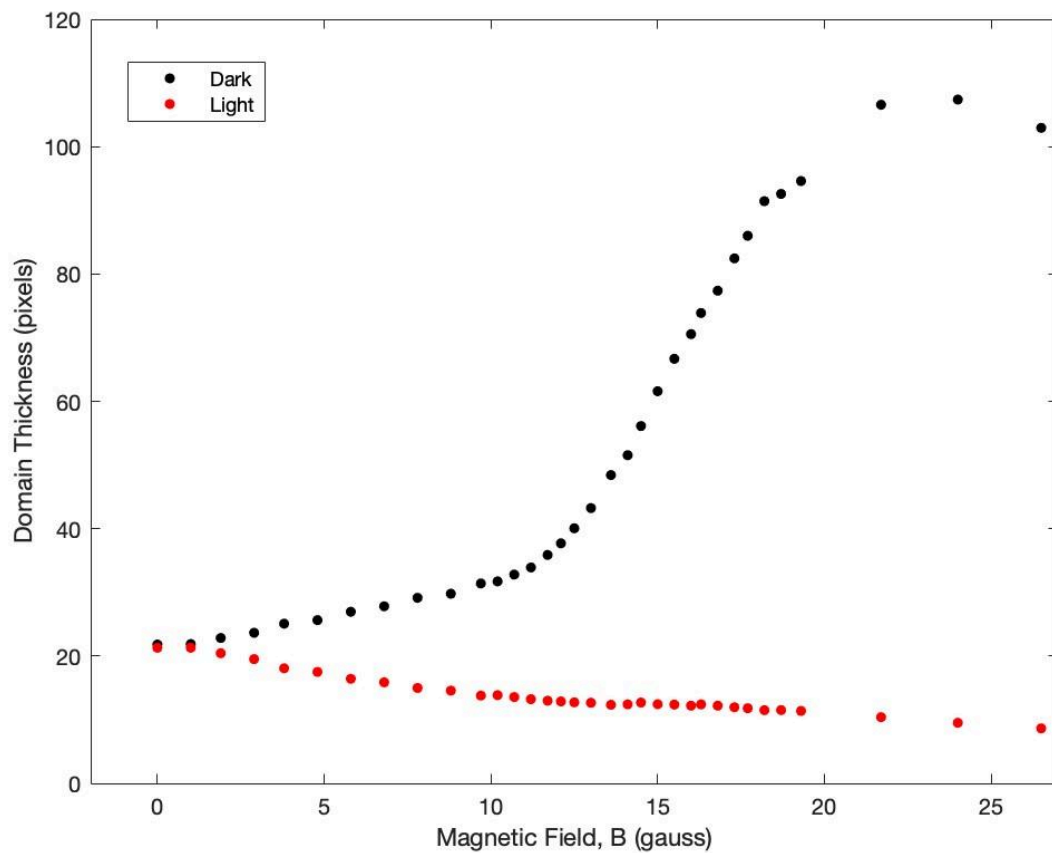


Figure 6. Domain Thickness versus magnetic field.

more random. I think the main thing driving this error is the imperfections in the image. Looking at images full of dense domains, the imperfections in the image don't really stand out, but when domains are sparse, these imperfections could make a significant difference in where the code counts many of the domains. The positive data set was affected by this effect more, and in the later images it can be seen that there is a gradient in the dark background color, which, if it were anywhere near the critical

threshold value, could cause false positives for small domains. One way to fix this would be to apply a binary filter and to convert every pixel in the dataset to either dark or light. Some analysis was done this way earlier in the experimentation, but since the error only occurs where the background domain thickness is known to be large, the greyscale thickness code is adequate for regions of interest where the domains are still dense. Additionally, the greyscale code is preferable as it the process for finding a critical value distinguishing between dark and light domains is completely transparent, while the binary images were converted to binary images from color images using ImageJ 2 where the process isn't as obvious or controllable. Next, I'll discuss the hysteresis data set. This consisted of taking images while changing the magnetic field in a linear

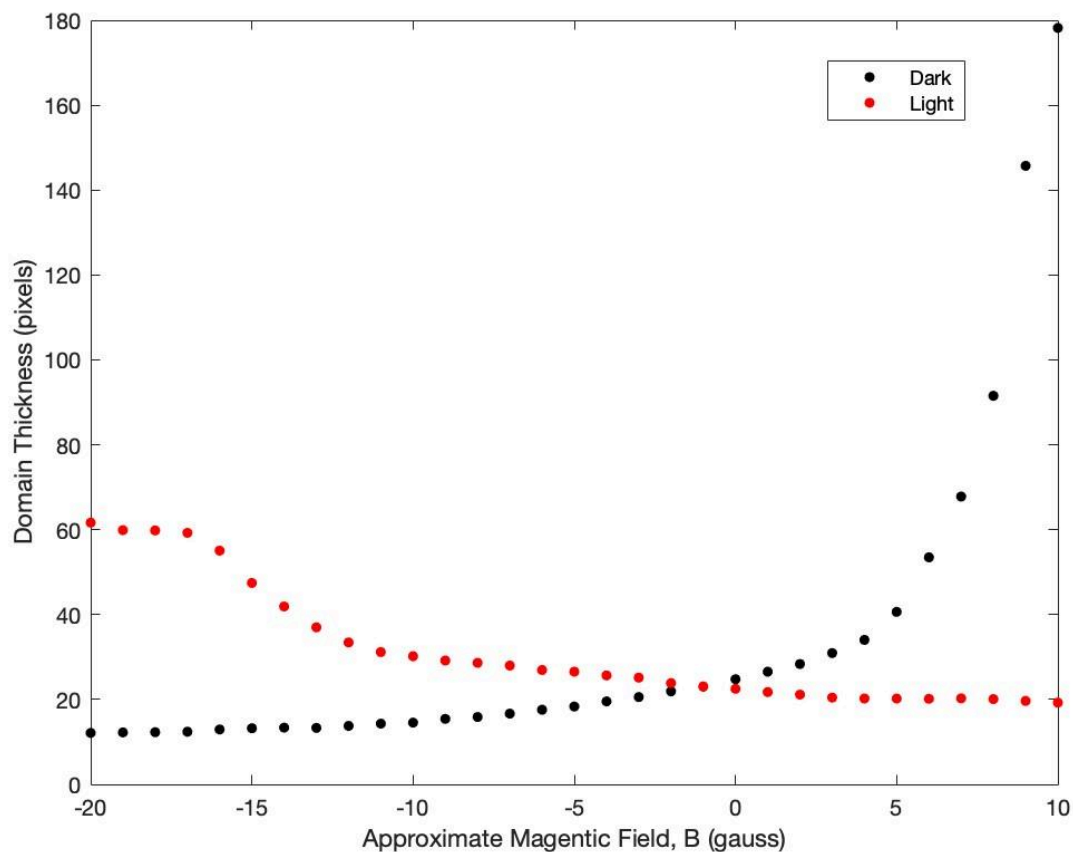


Figure 7. First leg of hysteresis data: 70 to -70 gauss.

fashion, similar to the step sets mentioned before, however in this case we started with a field around 70 gauss, dropped the field by about one gauss steps until we reached -70 gauss, then increased the field with one gauss steps to return to 70 gauss. This data was taken autonomously. Importantly, the magnetic domains are completely washed out between 30 and 40 gauss, so the most important data occurs between -40

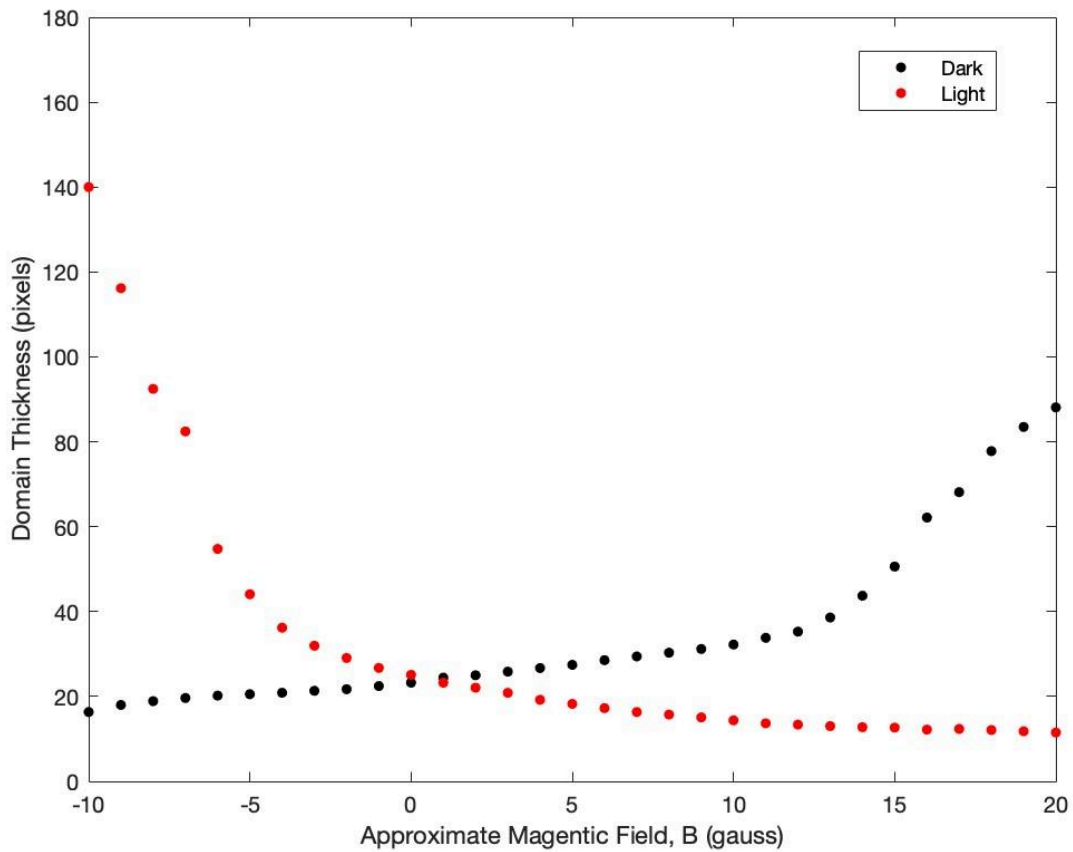


Figure 8. Second leg of hysteresis data: -70 to 70 gauss.

and 40 gauss, and images at higher magnitude fields are completely dark or light. As seen in the images, starting at the washed out field and decreasing to zero creates a similar pattern as the step data shown before. The step data shows a seemingly exponential increase in domain thickness for one color and a steady drop in thickness for the other as the field is changed in one direction; the hysteresis shows similar absolute results even as the field approaches zero (rather than starting at zero and approaching higher magnitude magnetic field values), and then displays, again, similar results after passing zero, albeit with a weaker increase in the background domain color thickness. All of this data shows that the domains behave similarly regardless of starting point; whether starting at zero or washed out field, the behavior is generally the same as the field is incremented in one direction. Another important note is that the hysteresis data set was prone to the same error as the step data; i.e., background thickness values become inaccurate as the sample becomes completely washed out. This is thought to be for the same reasons as before: imperfections in image. Again, the data before the washout is most important so it shouldn't affect the conclusions of the data significantly, and the graphs above omit the faulty thickness data. One difference between the data sets is that the hysteresis thickness data became unstable at different points in the magnetic field than the step data. The images also qualitatively looked very different, even though the domain thickness values are similar in the graphs at the same field values. This could've been caused by the fact that the step data was run after annealing the sample while the hysteresis data was, by nature, more of a quenched data set.

Grains

Computational Methods

While some data was collected during gradual changing of magnetic fields, some was collected with abrupt changes. One procedure, called quenching, involves rapidly changing the field to a large absolute value and then back down to zero (or any large jump in magnetic field all of the sudden). Another procedure, called annealing, involves quenching the sample exponentially in consecutively smaller steps. After quenching, the sample's domains are in a more disordered state, and after annealing, the domains are

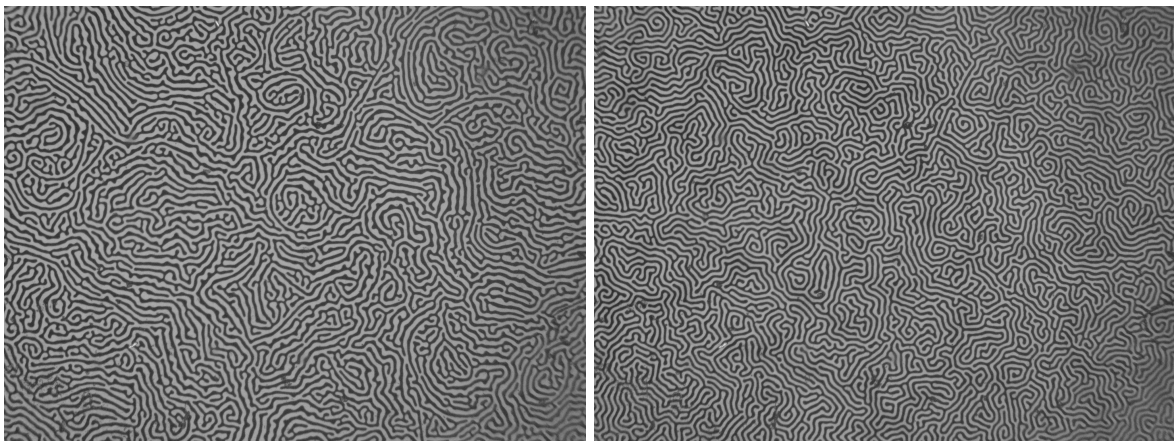


Figure 9. Quenched sample on the left, annealed sample on the right.

in a more ordered state. As can be seen in the images above featured in figure 9, the quenched sample and annealed sample show different types of patterns, with the patterns in the quenched sample tending to be larger and less orderly than the patterns in the annealed sample. These structures and the processes that create them resemble phenomena in material science. In a bulk material such as steel, analogous processes also named quenching and annealing result in different crystal structures in materials.

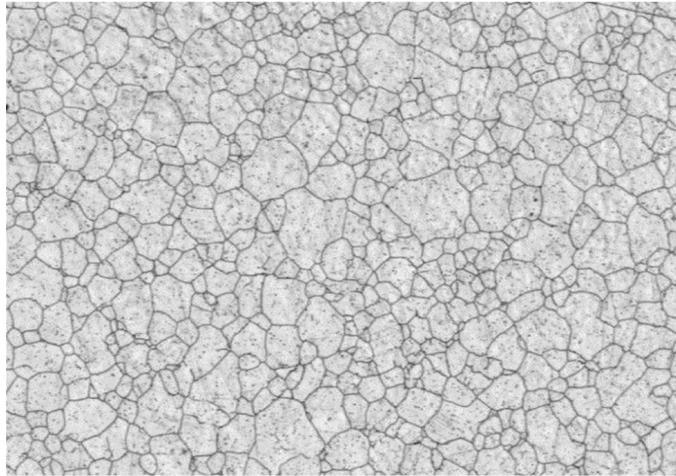


Figure 10. 100x magnification image of steel crystal grains (4).

The patterns and structures of these regions are called grains. Because of this similarity, I like to think of the patterns in the sample we analyzed as magnetic grains. Because these grains are clearly related to the previous environment the sample was exposed to, it is useful to quantify this effect as well. This was also accomplished in python. After trying some different methods, the seemingly most useful one is grain count. A number of successive domain lines in the same orientation would be one grain, so grain count just tallies these attributes into a single number. In the steel pictured above, and maybe even in the magnetic domain images above, it may be possible to physically count the grains present; however it is important to note the algorithm is an arbitrary method and doesn't count the actual number of visible grains in the sample. The algorithm works as follows: a specified number of gridlines are examined in the image. Along these gridlines, the length of each domain wall (found with the critical threshold value mentioned earlier) is examined and compared to that of its neighbors corresponding to the same color (dark lines compared to dark lines and light lines

compared to light lines) . If the domain thickness between adjacent lines is close enough (falls under some preset grain threshold), it can be assumed that the successive domain lines are in the same orientation, i.e. that they are parallel and thus a part of the same grain. The code is illustrated in figure 11. If two or more domain lines

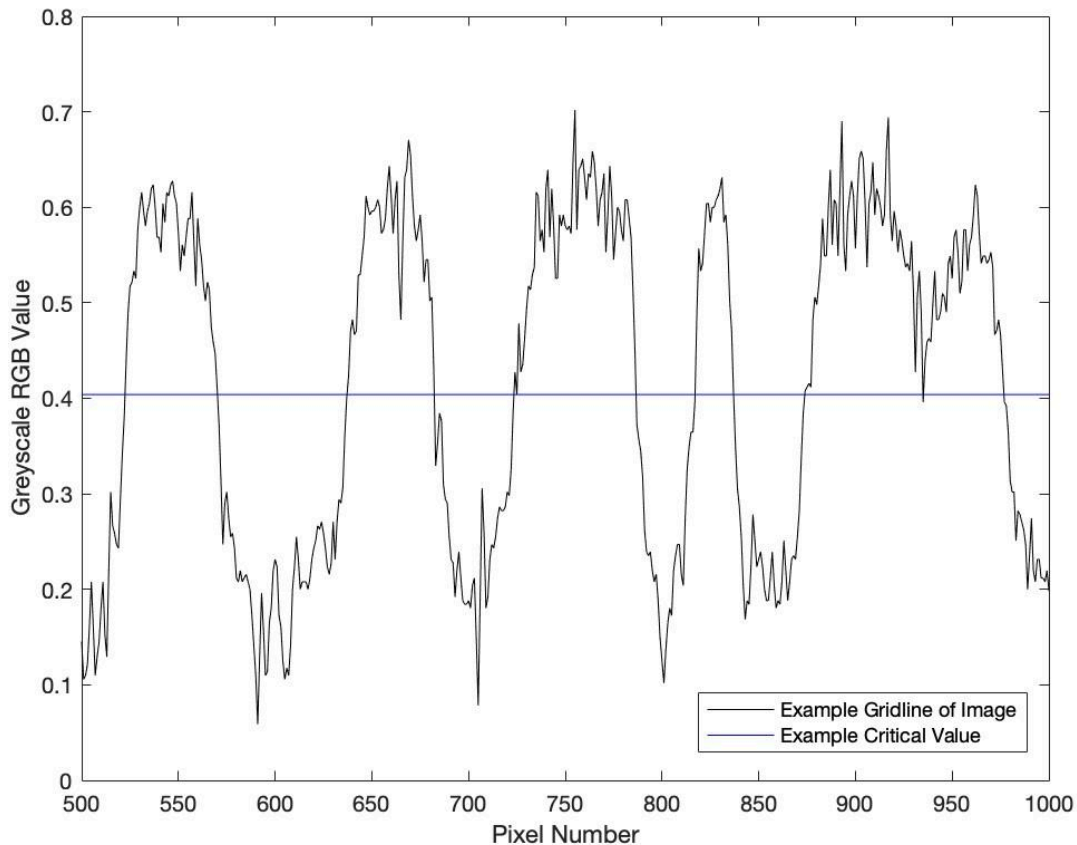


Figure 11. Graphic of grain code. This shows what the computer sees when it counts for grains down one of the gridlines. It works its way down the column or across the row and keeps track of how many pixels occur between two collisions with the critical value line. This shows an illustration of a section of a column run.

are deemed parallel by the code, a grain is counted. The code was calibrated with each set of data by testing all the runs with different preset grain thresholds and finding which one yields the largest difference between the quenched and annealed pictures (see Appendix A for graphs of calibration runs).

Results

The grain code was mainly run on one set of data, which I call square decay data. This data involved manipulating the magnetic field with a square wave multiplied by an exponential decay function, seen below. The images were taken when the

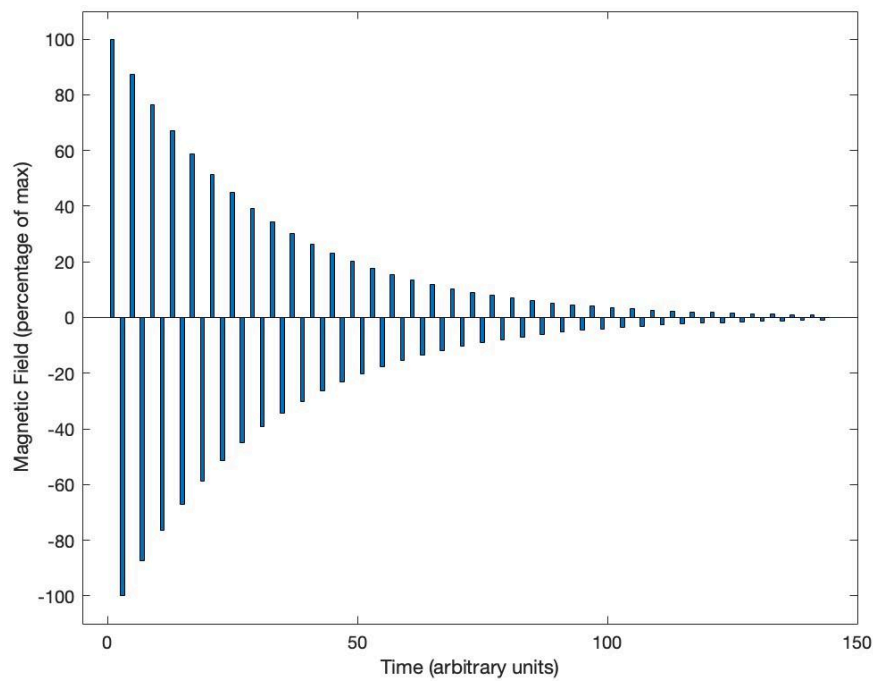


Figure 12. Magnetic field versus time. The plot above shows a similar square decay function to the one used in the experiment.

magnetic field was zero; so the field was abruptly changed to a non-zero value, held

there, and then abruptly changed back to zero between images. Looking at figure 12, this process anneals the sample going from left to right, so the sample goes from most quenched, on the left, to most annealed, on the right. As seen in figure 13, the number of grains increases from the quenched sample images to the more annealed sample images. Three different stages can be identified in this image: the quenched state, the transition state, and the annealed state. The specific data for the graph came from ten trials: five runs of the square decay function used starting with a positive peak, and five trials of the same function but with the negative peak first. Each positive run was paired with one negative run and the results were averaged; then the mean and standard deviation of these five averages were found and plotted. As can be seen in the

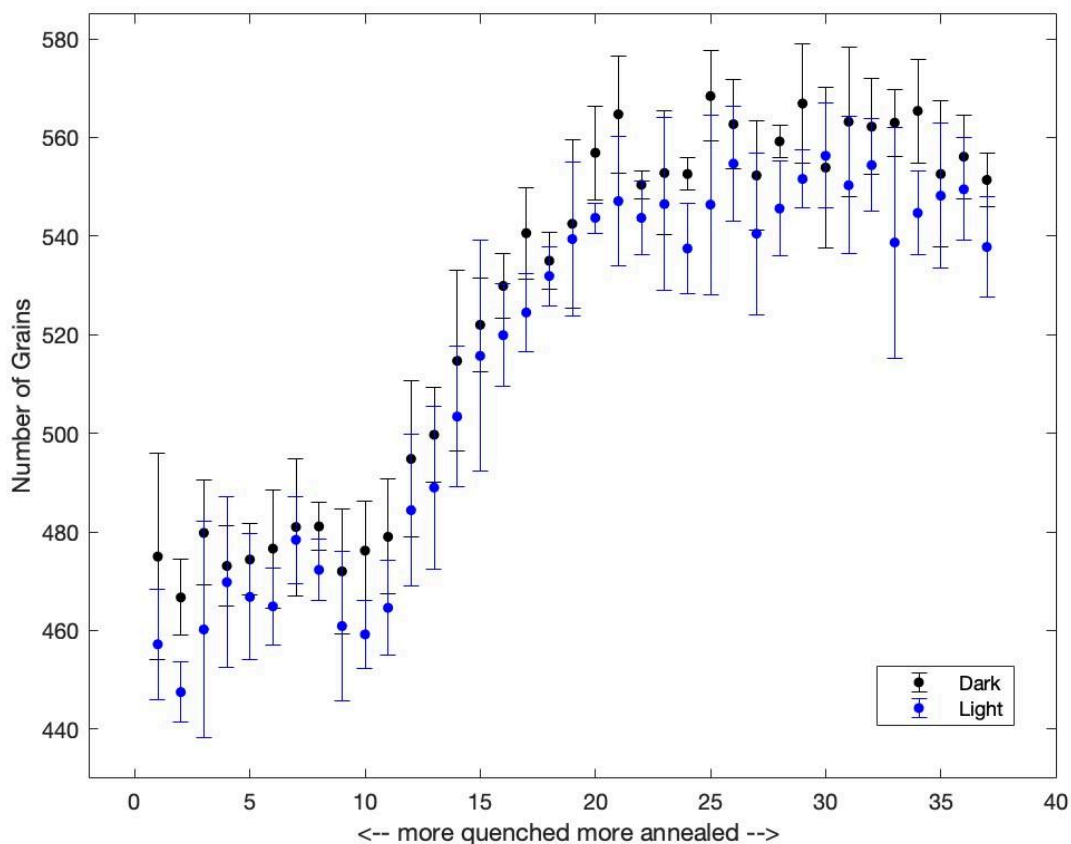


Figure 13. Number of grains versus sample state. This graph was created using a grain threshold of 25%, and with 20 vertical and 20 horizontal grid lines. The error bars represent the standard deviation of each data point.

graph, the error bars of the quenched region generally do not overlap with the error bars of the annealed region, indicating that there may be a significant difference between the regions. The sample has a memory of previous fields, and displays this in the patterns of domains that appear at zero field; the conclusion of the grain analysis is that this memory can indeed be quantified, and one way to quantify this difference is by counting grains using the python algorithm I created. This graph resembles an “S-curve” commonly seen in population growth graphs, where there is exponential growth until some ceiling is reached. This suggests that there is some maximum annealed, and thus, ordered, state of the sample.

Conclusion

The magnetic domains of a Bi:YIG sample were successfully imaged using the Fourier Effect, and I was successfully able to quantify different attributes of these magnetic domain images with algorithms written in python. There are boundless future research opportunities with these types of samples and methods; one that could be used to improve the data I collected is running more step data trials to get averaged results, as well as running multiple step data trials after quenching the sample to compare the quenched and annealed results and see if there are any notable differences. Another interesting experiment would be to use a different material with all

else held constant, i.e. running the same trias and algorithms and comparing the data between the two samples. Some possible real world applications of my findings include using the grain analysis to estimate a Bi:YIG sample's external conditions with images of the sample at a later time, and using the thickness analysis to estimate the magnetic field present when a particular image was taken of a Bi:YIG sample.

Appendices

Appendix A. Calibration curves for red sample grain analysis.

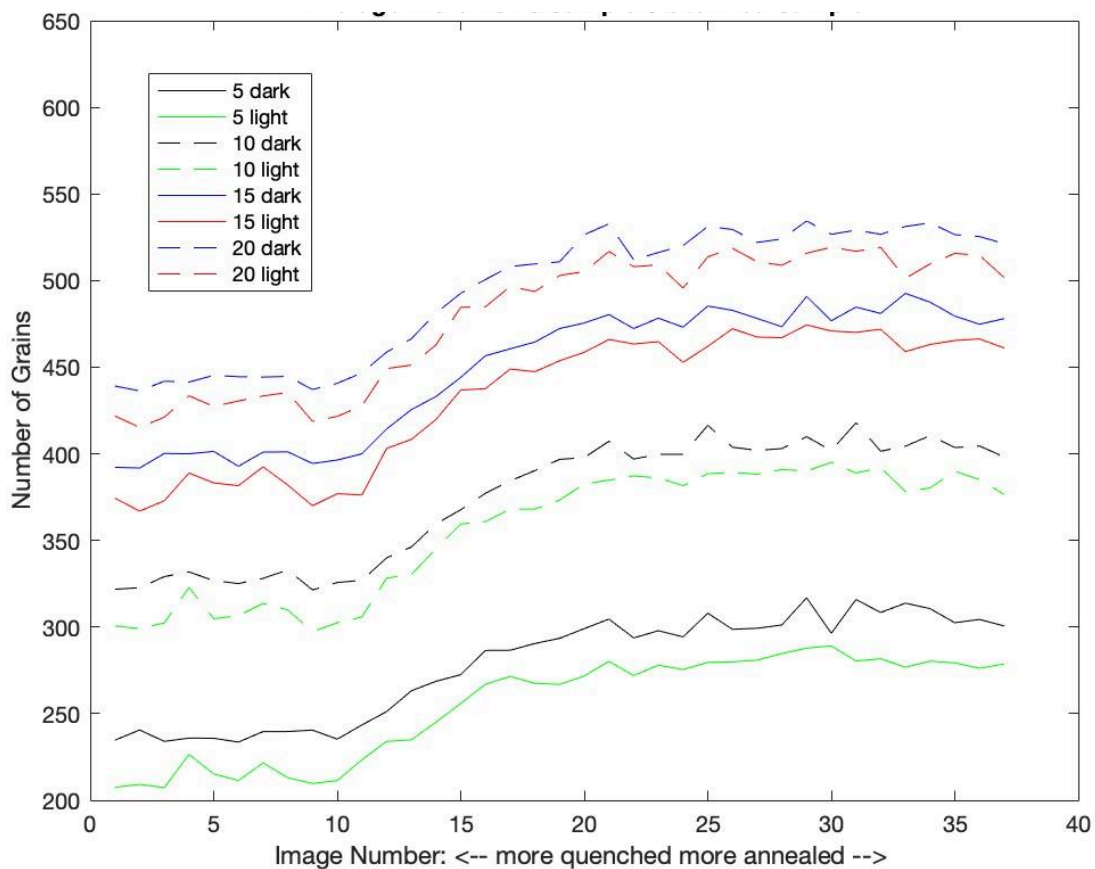


Figure 14. Number of grains versus sample state, grain thresholds 5 to 20%.

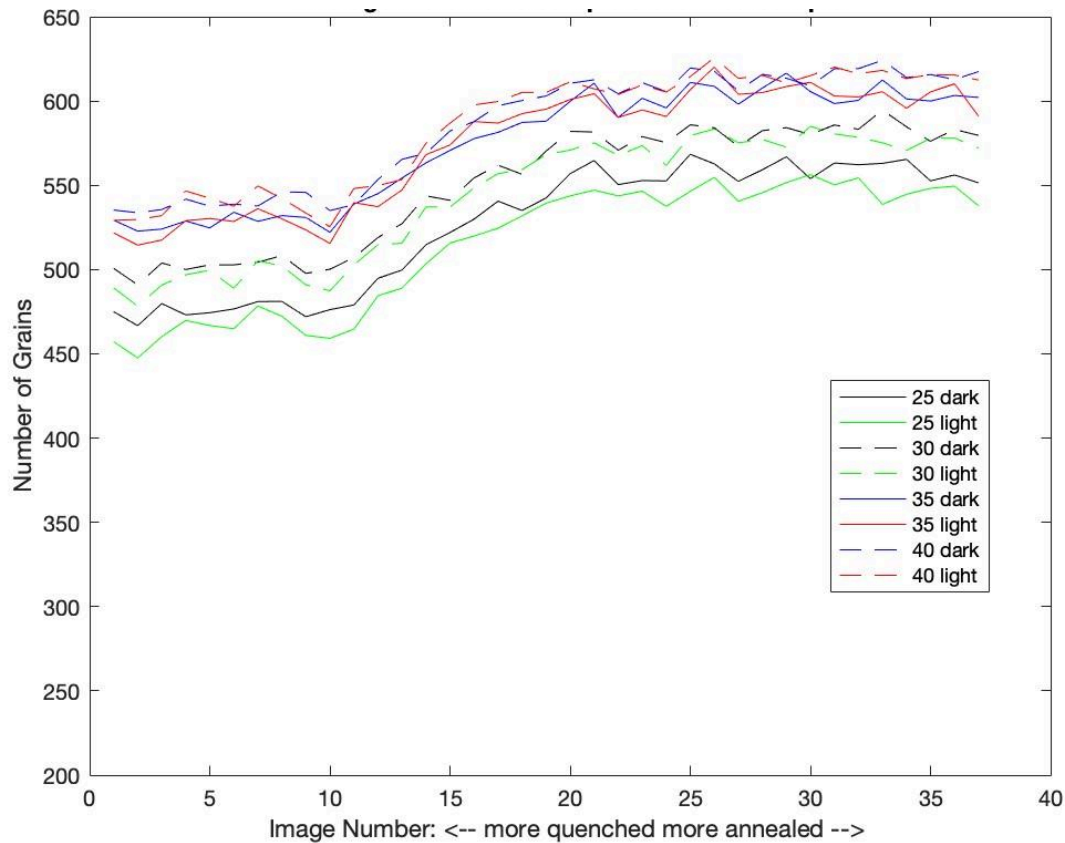


Figure 15. Number of grains versus sample state, grain thresholds 25 to 40%.

References

- (1) Wikipedia, *Magnetic Domains*. [https://en.wikipedia.org/wiki/Magnetic_domain#Magneto-optic_Kerr_effect_\(MOKE\)](https://en.wikipedia.org/wiki/Magnetic_domain#Magneto-optic_Kerr_effect_(MOKE))
- (2) Wikipedia, *Faraday Effect*. https://en.wikipedia.org/wiki/Faraday_effect
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- (4) Clemex, *Grain Size Analysis in Steel*. <https://clemex.com/analysis/grain-size-in-steel-2/>